

Supplementary Information

Operating regimes and materials limits of nonlinear magnon dynamics in Co-doped YIG

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This document provides the numerical protocols, parameter tables, derivations, and reproducibility checklist for the main text. Supplement 8 documents the completed 310 K static-drive frequency scan used for Fig. 10 and Table 3. Supplement 9 documents the thermal-start population protocol, the high-occupation quantum-trajectory method, detailed-balance and transition-resolved checks, and the reset-and-read transient phase-response convergence audit. Supplement 10 gives the exact quadratic anisotropy decomposition and clarifies the scope of the laboratory mapping of the effective pair-drive protocols.

Supplement 1. Numerical models, parameters, and reproducibility

S1.1 Common open-system convention

The zero-temperature audit calculations use a single bosonic mode with annihilation operator $\hat{a}$, number operator $n = \hat{a}^\dagger \hat{a}$, and a zero-temperature single-quantum loss channel. The finite-temperature calculations of Supplements 8–9 add the corresponding thermal absorption channel explicitly. The zero-temperature audit master equation is

$$d\rho/dt = -i[H(t), \rho] + \kappa D[\hat{a}]\rho, \quad D[\hat{a}]\rho = \hat{a}\rho\hat{a}^\dagger - (1/2)\{\hat{a}^\dagger\hat{a}, \rho\}.$$

The damping convention is $\kappa = 2\alpha\Omega$. With this convention the coherent amplitude decays at the Gilbert rate $\alpha\Omega$ because the dissipator contributes $-(\kappa/2)\langle\hat{a}\rangle$ to $d\langle\hat{a}\rangle/dt$.

The calculation scripts use $\Omega = 1$, $U_{\text{audit}} = 0.55$, $\eta = 1.2$, $\alpha = 0.01$, $\kappa = 0.02$, a 50-state Fock basis, normalized time Ωt from 0 to 100, and 3001 requested output times. QuTiP integrates adaptively between these output times.

S1.2 Hamiltonian conventions

The matched audit calculations use the nonlinear Hamiltonian $H_0 = \Omega \hat{a}^\dagger \hat{a} + (U_{\text{audit}}\Omega/2) \hat{a}^\dagger{}^2 \hat{a}^2$. The main text writes the Kerr term as $U\Omega \hat{a}^\dagger{}^2 \hat{a}^2$ in Eq. (4), so the two coefficients are related by $U = U_{\text{audit}}/2$, and it is U that enters the criterion of Section 8 of the main text.

Here U_{audit} is the dimensionless Kerr coefficient of the audit normalization; the corresponding nonlinear frequency scale in the convention of Eq. (4) of the main text is $K = U\Omega$ with $U = U_{\text{audit}}/2$. Two numerical pair-drive protocols are compared below. The static effective Hamiltonian coefficient is $\eta\Omega/4$. The resonant Hamiltonian coefficient is $(\eta\Omega/2)\sin(2\Omega t)$, and the corresponding equation-of-motion coefficient is twice that

value because $[\hat{a}, \hat{a}^{\dagger 2}] = 2\hat{a}^{\dagger}$. In these numerical protocols η is the dimensionless effective pump amplitude. The derivative-based sweep non-adiabaticity $\eta_{\text{NA}} = |\text{d}\Omega/\text{d}t|/\omega^2$ is a distinct quantity unless a specific microscopic laboratory modulation supplies a calibration between them. Supplement 10 gives the exact quadratic anisotropy decomposition and explains why a scalar $\Omega(t)$ alone does not determine the pair coefficient. The static calculation remains a separate effective benchmark.

- Static effective pair coupling: $H_{\text{drive}} = (\eta\Omega/4)(\hat{a}^{\dagger 2} + \hat{a}^2)$.
- Explicit resonant modulation: $H_{\text{drive}}(t) = [\eta\Omega \sin(2\Omega t)/2](\hat{a}^{\dagger 2} + \hat{a}^2)$.

Important limitation. Two Kerr normalizations appear in the calculations of this work. A term Un^2 is not numerically identical to $(U\Omega/2)\hat{a}^{\dagger 2}\hat{a}^2$. Since $\hat{a}^{\dagger 2}\hat{a}^2 = n(n-1)$, changing conventions also changes the effective linear frequency and the coefficient multiplying the anharmonicity. The symbol U must therefore not be transferred between those calculations without an explicit normalization conversion. For the two conventions actually used in this work the conversion is a factor of two: with $\Omega = 1$ the matched-audit term $(U_{\text{audit}}\Omega/2)\hat{a}^{\dagger 2}\hat{a}^2$ equals $(U_{\text{audit}}/2)\hat{a}^{\dagger 2}\hat{a}^2$, so a nominal $U_{\text{audit}} = 0.55$ carries half the anharmonicity of a nominal $U = 0.55$ in the convention of Eq. (4). Every calculation reported here uses the audit normalization — the drive-protocol audit of Supplement 3, the bridge of Supplement 4, the pair-phase control of Supplement 5 and the 310 K frequency scan of Supplement 8 are all integrated with $(U_{\text{audit}}\Omega/2)\hat{a}^{\dagger 2}\hat{a}^2$ — and all convert as $U = U_{\text{audit}}/2$. The direct normalization $U\hat{a}^{\dagger 2}\hat{a}^2$ appears only in auxiliary runs retained in the reproducibility archive, which are not used for any quoted value, and in the 1000-pump-period (=500-carrier-period) calculation of Fig. 3 of the main text, whose $U = 0.1$ is therefore already an Eq. (4) value. The conversion is not a labelling convention only: the mean-field balance of Section 5.3 of the main text predicts a saturated occupation $\eta/(4U)$, which at the representative resonant point ($\eta = 1.2$, $U_{\text{audit}} = 0.55$, hence $U = 0.275$) gives 1.09 against the 1.213 returned by the full Lindblad calculation of Supplement 4, an agreement to 11%, whereas reading U_{audit} directly as U would predict 0.545 and miss by a factor of two. The computed dynamics therefore selects the same conversion as the algebra.

S1.3 Observables

Quantity	Definition	Interpretation
$ \nu ^2$	Reduced Bogoliubov occupation	Mean occupation estimate, not P_1
$\langle n \rangle$	$\text{Tr}(\rho \hat{a}^{\dagger} \hat{a})$	Full quantum mean occupation
P_j	$\langle j \rho j \rangle$	Probability of exactly j quanta
$\langle \hat{a} \hat{a} \rangle$	$\text{Tr}(\rho \hat{a} \hat{a})$	Anomalous pair coherence
$\langle (-1)^n \rangle$	Parity expectation	Even/odd-sector imbalance

S1.4 Numerical convergence and provenance

Convergence statements are supported only when directly tied to an archived script and verified output file. Claims not documented in the archive — including any yield, throughput, or Monte-Carlo figures — are not part of the validated results of this paper.

The 40- and 100-level calculations demonstrate that the upper Fock cutoff was not populated for the stated static effective-drive cases. This is a basis-convergence check; it is not evidence that the physical state is an exact two-level state.

Supplement 2. Relation between the Lindblad and Bogoliubov descriptions

S2.1 Exact linear structure and noise term

For $U = 0$, the Hamiltonian is quadratic and the Heisenberg-Langevin equation has the standard form

$$d\hat{a}/dt = -i[\hat{a}, H(t)] - (\kappa/2)\hat{a} + \sqrt{\kappa} \hat{a}_{\text{in}}(t).$$

The input-noise term is required in the open-system operator equation. A deterministic reduced equation that keeps only the damping term does not preserve the canonical hyperbolic norm of a unitary Bogoliubov transformation.

S2.2 Mean-field nonlinear closure

The quartic Kerr interaction generates higher operator moments, so the dynamics do not close exactly on two complex amplitudes $u(t)$ and $v(t)$. The reduced equations used in the audit replace the operator interaction by a state-dependent frequency shift $\Delta = U_{\text{audit}}\Omega|v|^2$. This is the Hartree closure of the audit Hamiltonian, in which the Kerr term is $(U_{\text{audit}}\Omega/2)\hat{a}^\dagger{}^2\hat{a}^2$ and the shift is therefore twice the Kerr coefficient times the occupation; in the convention of Eq. (4) of the main text the same quantity reads $2U\Omega|v|^2$ with $U = U_{\text{audit}}/2$, so the two statements agree. It is an approximation, not an operator identity.

The matched reduced equations are integrated with the same resonant modulation as the corresponding Lindblad calculation. The coefficient appearing in the reduced (u, v) equations is $G(t) = \eta\Omega \sin(2\Omega t)$, twice the pair coefficient $\eta\Omega \sin(2\Omega t)/2$ that appears in the Hamiltonian, because $[\hat{a}, \hat{a}^\dagger{}^2] = 2\hat{a}^\dagger$ introduces a factor of two on passing to the equations of motion. The two therefore describe the same drive. Agreement should therefore be assessed for mean occupation and phase trends, not for the complete Fock distribution or all higher moments.

S2.3 Damped norm diagnostic

For the deterministic reduced equations with equal damping $-\alpha\Omega u$ and $-\alpha\Omega v$, direct differentiation gives

$$d/dt (|u|^2 - |v|^2) = -2\alpha\Omega (|u|^2 - |v|^2),$$

and hence $|u|^2 - |v|^2 = e^{-2\alpha\Omega t}$ for $u(0)=1$ and $v(0)=0$. At $\alpha = 0.01$ and $\Omega t = 100$ this equals $e^{-2} = 0.1353352832$. The numerical audit reproduces it with a maximum deviation of 2.4×10^{-9} from the analytic decay over the whole interval (Figure S1). This diagnostic confirms the integrator; it must not be interpreted as a transition from a hyperbolic to a Euclidean quantum-state normalization.

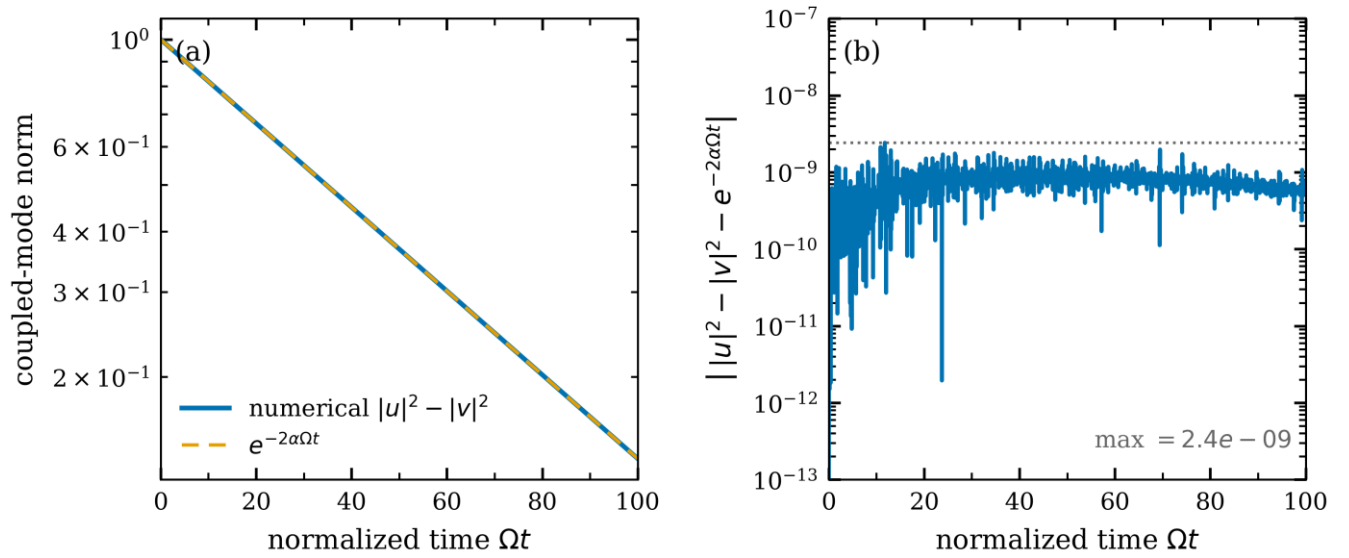


Figure S1. Damped coupled-mode norm diagnostic. (a) The hyperbolic norm $|u|^2 - |v|^2$ of the reduced equations (solid) against the analytic decay $\exp(-2\alpha\Omega t)$ (dashed); the two are indistinguishable on this scale. (b) The absolute deviation between them, which stays below 2.4×10^{-9} over the whole interval. The quantity plotted is a diagnostic of the integrator and of the loss term in the reduced equations; it is not a state normalization, and the decay it shows is not a physical loss of probability.

Supplement 3. Drive-protocol audit

The purpose of this audit is to isolate the effect of the pair-drive implementation while holding the matched Kerr convention, damping convention, basis size, initial state, and time interval fixed.

Protocol	Late mean $\langle n \rangle$	Late mean P_0	Late mean P_1	Late mean $P(n \geq 2)$	Final parity
Static effective	0.11696	0.91857	0.04952	0.03192	0.89352
Resonant modulation	1.21332	0.31152	0.33347	0.35500	0.15432

The resonantly modulated protocol produces a late-time mean occupation approximately 10.4 times larger than the matched static effective protocol. Thus nominally identical labels (U, η, α) do not identify the same physical calculation when the drive implementation differs.

The matched static audit gives a late-time mean occupation of 0.11696. An auxiliary calculation in the reproducibility archive that carries the same parameter labels but uses a different drive implementation and a different Kerr normalization gives 0.0404; it is not used for any value quoted in the main text.

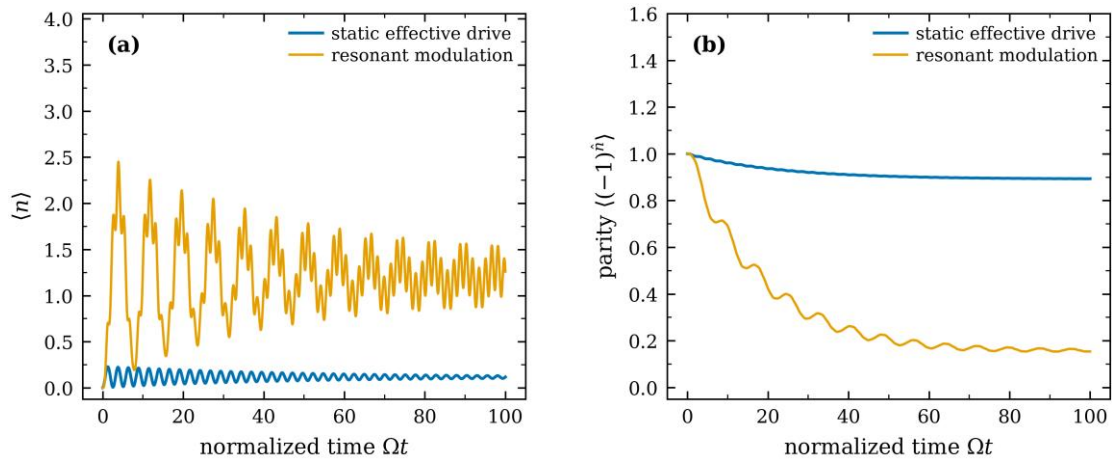


Figure S2. Protocol audit of the mean occupation. The static effective and explicitly resonant pair drives produce distinct bounded occupation scales under the matched Kerr convention.

Supplement 4. Matched Bogoliubov–Lindblad bridge

Under the resonantly modulated protocol at $U_{\text{audit}} = 0.55$ (that is, $U = 0.275$ in the convention of Eq. (4) of the main text), $\eta = 1.2$, $\alpha = 0.01$ and $\kappa = 2\alpha\Omega$, the reduced calculation gives a late-time mean $|v|^2 = 1.11640$ with standard deviation 0.49405. The Lindblad calculation gives a late-time mean $\langle n \rangle = 1.21332$ with standard deviation 0.22237. Their ratio is 0.9201.

The reduced and full Lindblad descriptions therefore agree on the time-averaged occupation at the 8% level, which is the level expected from the mean-field closure. It does not imply agreement of the instantaneous trajectories: the maximum pointwise difference is 2.56 and the RMS difference is 0.72.

The Lindblad state is not a one-quantum Fock state. At the representative resonant point the late-time populations are broadly distributed, with mean $P_0 = 0.3115$, mean $P_1 = 0.3335$, and mean $P(n \geq 2) = 0.3550$. The maximum P_1 observed in the run is 0.38175.

S4.1 Parity structure

The Hamiltonian terms $\hat{a}^\dagger \hat{a}$, $\hat{a}^\dagger{}^2 \hat{a}^2$, and $\hat{a}^\dagger{}^2 + \hat{a}^2$ commute with the number-parity operator $(-1)^n$. Starting from the vacuum, the unitary sector therefore populates only even Fock states. Odd populations arise through the single-quantum jump operator $\hat{a}$.

This establishes that P_1 is not coherently addressed by the pair Hamiltonian. The present calculation does not provide deterministic coherent preparation of $|1\rangle$. A rigorous universal upper bound on transient P_1 for every driven-dissipative parameter choice is not claimed here.

S4.2 Anomalous-correlator convention

Candidate reduced correlator	Complex RMS	Magnitude RMS	Magnitude correlation	Mean phase shift	Phase locking
uv	1.7600	0.4817	0.3411	-2.2474 rad	0.0199
$u v^*$	0.6702	0.4817	0.3411	0.0145 rad	0.8122
-uv	1.7535	0.4817	0.3411	0.8941 rad	0.0199
$-u v^*$	2.3923	0.4817	0.3411	-3.1271 rad	0.8122

The convention $u v^*$ gives the best complex agreement with the Lindblad anomalous correlator $\langle \hat{a} \hat{a} \rangle$. Its near-zero mean phase offset and phase-locking resultant 0.812 indicate that the reduced model captures the pair-phase trend more reliably than the pair-correlation magnitude.

The magnitude RMS difference remains 0.482 and the magnitude correlation is 0.341. Therefore the reduced pair amplitude is only qualitative.

Supplement 5. Pair-phase control and dynamical step response

S5.1 Stationary transfer of the external drive phase

For the resonantly modulated protocol $G_\varphi(t) = \eta \Omega \sin(2\Omega t + \varphi_d)$, five Lindblad calculations were performed at $\varphi_d = 0, \pi/4, \pi/2, 3\pi/4$, and π . The late-time rotating-frame pair phase was defined as $\Phi_{\text{pair}}(t) = \arg(\hat{a} \hat{a}) + 2\Omega t$. The fitted relation is

$$\bar{\Phi}_{\text{pair}} = -1.5589 - 0.99984 \varphi_d, \quad R^2 = 0.9999993,$$

with an RMS fit deviation of 9.3×10^{-4} rad. Thus the stationary pair phase follows the imposed pump phase nearly one-to-one, with the minus sign fixed by the adopted phase convention. Across the same phase range, the late-time mean occupation changes only from 1.2133 to 1.2252 and the mean pair-correlation amplitude changes from 1.1035 to 1.1309. The circular phase standard deviation remains 0.225-0.232 rad. These are temporal variations of the ensemble-averaged correlator, not a quantum-trajectory phase-diffusion measurement.

S5.2 Dynamical phase-step protocol

A phase step breaks the time-translation equivalence of the stationary five-phase scan and therefore tests genuine dynamical control. The Hamiltonian discontinuity at $\Omega t = 100$ is integrated as two separate mesolve segments: the final density matrix of the zero-phase segment is used as the initial state of the post-step segment, while absolute time is retained in the rotating-frame transformation.

The analysis uses the complex rotating-frame correlator $z(t) = \langle \hat{a}\hat{a} \rangle(t) e^{2i\Omega t}$. The response time is extracted from the complex distance $|z(t) - z_\infty|$, avoiding phase unwrapping when the amplitude passes through zero during large phase steps. This convention is documented in the archived script `phase_step_test_v2.py`.

For phase steps $\varphi_{\text{step}} = \pi/8, \pi/4, \pi/2$, and π , the steady-state transfer ratios $-\Delta\Phi/\varphi_{\text{step}}$ are 0.99477, 1.00045, 0.99764, and 0.99688, respectively. The corresponding complex-fit response times are 14.485, 19.957, 26.120, and 21.734 Ω^{-1} , all shorter than the damping time $(\alpha\Omega)^{-1} = 100 \Omega^{-1}$. The response is therefore drive-controlled rather than set by slow damping.

The phase-only interpretation has a finite operating range. For $\pi/8$ and $\pi/4$ steps the minimum pair-amplitude ratios during the transient are 0.703 and 0.451. For $\pi/2$ and π steps they fall to 0.005 and 0.015, so the pair correlator passes close to the origin and its instantaneous phase is temporarily ill-defined. The corresponding peak occupations rise to 2.168 and 2.219 times their pre-step values. Large phase steps therefore involve strong amplitude-phase coupling rather than a pure phase rotation.

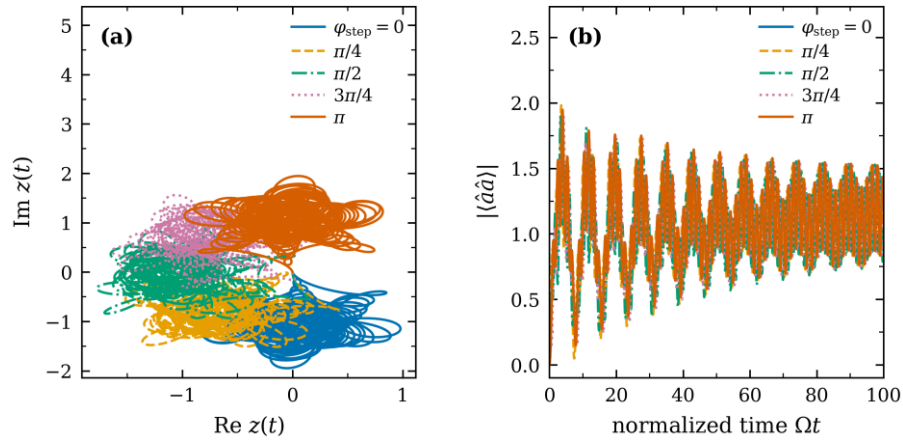


Figure S3. Complex-plane step response of the anomalous pair mode. The pump phase is switched at $\Omega t = 100$ from zero to $\varphi_{\text{step}} = 0, \pi/8, \pi/4, \pi/2$, and π . Left: trajectories of $z(t) = \langle \hat{a}\hat{a} \rangle \exp(2i\Omega t)$ in the complex plane. Right: pair-correlation amplitude across the step. The final phase transfer is nearly one-to-one for all tested steps. For $\pi/2$ and π , however, the pair amplitude collapses close to zero, so the instantaneous phase is temporarily ill-defined and phase and amplitude are not dynamically decoupled.

Supplement 6. Origin of the energy scale and what is not claimed

The 22–24 aJ switching-event estimate is obtained from the experimentally reported absorbed-energy density multiplied by the assumed $20 \times 20 \times 10 \text{ nm}^3$ cell volume. It is not calculated from $\langle n \rangle$, $|v|^2$, $\hbar\Omega$, or the dimensionless Kerr coefficient U .

The normalized calculations demonstrate bounded dynamics for specified model protocols; by themselves they fix no material scale. The microscopic mapping from U to a material-specific YIG:Co Kerr coefficient is supplied separately, in Section 5.1 of the main text, and gives $U = \omega_a/(2N_{\text{eff}}\Omega)$ with the net macrospin content $N_{\text{eff}} = M_s V/(g\mu_B)$, equivalently a Kerr coefficient $K = \hbar\gamma^2 K_u/(M_s^2 V)$ expressed in measurable quantities alone. It is that mapping, and not these normalized runs, which establishes that the values of U used here are not realized by any normal mode of a uniformly magnetized cell of the assumed size.

Throughput, passive-cooling, readout-energy, fabrication-yield, and multi-cycle-logic figures require a separate device-level model and are not validated outputs of the present single-mode simulations. No such figures are claimed in this paper.

The software archive contains: the five-phase stationary test scripts, the phase-step response scripts, the publication-figure generator, the finite-temperature frequency-scan scripts and results (50 GHz–10 THz), CSV and JSON numerical summaries, and this Supplementary Information document.

All calculation scripts are archived at the project repository: <https://github.com/alexandermtishin-art/Nonlinear-magnon-dynamics-in-Co-doped-YIG-four-operating-regimes>. The archive contains the Python/QuTiP scripts for each figure, processed CSV outputs, and a SHA256 checksum manifest.

Supplement 7. Reproducibility checklist

1. Record the exact Hamiltonian, including the coefficient and convention of the Kerr term.
2. State whether the pair coefficient is static in the calculation frame or explicitly modulated in time.
3. Define η independently from any coefficient that is merely labelled by η .
4. Use $\kappa = 2\alpha\Omega$ when $\alpha\Omega$ is intended as the amplitude-decay rate.
5. Distinguish final-time values, late-time means, maxima, and time-averaged standard deviations.
6. Distinguish $|v|^2$, $\langle n \rangle$, P_1 , P_0+P_1 , and $P(n \geq 2)$.
7. Report basis-cutoff convergence separately from physical low-level confinement.
8. Archive the code version, package versions, random seeds if applicable, and all generated CSV files.
9. Do not infer a Fock-state fidelity from a mean occupation close to one.
10. Do not infer phase diffusion from a single ensemble-averaged Lindblad trajectory.

Supplement 8. Finite-temperature frequency scan at 310 K

S8.1 Model and numerical protocol

The room-temperature frequency scan uses the thermal Lindblad equation with the same single-mode operators as above, but with both downward and upward bath channels:

$$L_{\downarrow} = [\kappa(\bar{n}_{\text{th}}+1)]^{1/2} \hat{a}, \quad L_{\uparrow} = (\kappa\bar{n}_{\text{th}})^{1/2} \hat{a}^{\dagger}, \quad \bar{n}_{\text{th}} = [\exp(hf/k_{\text{B}}T)-1]^{-1}.$$

The scan Hamiltonian is $H = \Omega\hat{n} + (U_{\text{audit}}\Omega/2)\hat{a}^{\dagger 2}\hat{a}^2 + (\eta\Omega/4)(\hat{a}^{\dagger 2}+\hat{a}^2)$, evaluated in normalized units $\Omega = 1$, with $U_{\text{audit}} = 0.55$ (that is $U = 0.275$ in the convention of Eq. (4) of the main text), $\eta = 1.2$, $\alpha = 10^{-3}$, $\kappa = 2\alpha\Omega$, $T = 310$ K, vacuum initial state, 20 pump periods $T_p = \pi/\Omega$ ($=10$ carrier periods $T_c = 2\pi/\Omega$), and 10 requested output points per pump period. The Fock basis is not fixed: it is chosen at each frequency from the occupancy actually reached, and every point is repeated at 1.2 times that size as a convergence test. Frequencies are 50, 75, 100, 150, 200, 300, 500, and 750 GHz, followed by 1, 1.5, 2, 3, 4, 5, 6, 7, 8, 9, and 10 THz. Basis convergence is demanding only at the low-frequency end, where the thermal occupation of the undriven mode is large: $\bar{n}_{\text{th}} = 128.7$ at 50 GHz. The run starts from vacuum and covers 20 cycles, so that $\kappa t = 0.126$ and the driven mode reaches $\langle n \rangle = 15.48$ rather than its equilibrium value; the basis selected there is $N = 209$, with the convergence repeat at $N = 250$. The measured population of the ten highest levels is 2.5×10^{-6} and the largest relative change of any reported observable between the two basis sizes is 4.6×10^{-5} . The basis falls to $N = 121$ at 100 GHz, 77 at 200 GHz and 45 at and above 700 GHz, where the top-ten population is 4×10^{-10} or smaller. These are direct convergence tests of the driven state, not a priori estimates from a thermal distribution of the same mean; the archived per-frequency values are in the scan output file.

Convention. This scan uses the same audit normalization as Supplements 3–5, $(U_{\text{audit}}\Omega/2)\hat{a}^{\dagger 2}\hat{a}^2$, and converts to the main-text coefficient as $U = U_{\text{audit}}/2$. A companion scan retained in the reproducibility archive uses the direct normalization $U\hat{a}^{\dagger 2}\hat{a}^2$ and a fixed 140-level basis; its numbers differ (at 6 THz $P_0+P_1 = 0.9593$ and $\langle n \rangle = 0.1643$ against 0.9254 and 0.2452 here) and are not used anywhere in the main text. Results from the two conventions must remain explicitly labelled.

For each frequency the archived output records the final mean occupation $\langle n \rangle$, the low-manifold population P_0+P_1 , the leakage $P(n \geq 2)$, the anomalous pair amplitude $|\langle \hat{a}\hat{a} \rangle|$, the normalized pair measure $|\langle \hat{a}\hat{a} \rangle|/(2\langle n \rangle+1)$, and the population of the ten highest basis levels.

S8.2 Completed 6–10 THz results for $U_{\text{audit}} = 0.55$

The final high-frequency points of the completed scan are:

f (THz)	$\bar{n}_{\text{th}}$	Final $\langle n \rangle$	P_0+P_1	$P(n \geq 2)$	$ \langle \hat{a}\hat{a} \rangle $	Normalized pair
6	0.652875	0.245215	0.925418	0.074582	0.393312	0.263892
7	0.511353	0.228658	0.926351	0.073649	0.397639	0.272857
8	0.408083	0.216617	0.926733	0.073267	0.401108	0.279862
9	0.330225	0.207562	0.926841	0.073159	0.403908	0.285422
10	0.270071	0.200580	0.926811	0.073189	0.406184	0.289891

Across 6–10 THz the low-manifold population rises only from 0.9254 to 0.9268, while the final mean occupation falls from 0.2452 to 0.2006. The absolute pair amplitude rises slightly from 0.393 to 0.406 and the normalized pair measure increases from 0.2639 to 0.2899, consistent with the table above, since $|\langle \hat{a}\hat{a} \rangle|/(2\langle n \rangle+1)$ reproduces the tabulated column to six digits at every frequency. The population in the highest ten basis levels is below 10^{-118} at every point, so the $N = 140$ truncation is not limiting these high-frequency results.

The scan does not show a sharp transition at 6 THz. Recovery of low-Fock confinement is progressive across the several-THz range: P_0+P_1 is 0.9079 at 3 THz, 0.9187 at 4 THz, 0.9233 at 5 THz, and 0.9254 at 6 THz. Accordingly, 6 THz should be treated as a representative room-temperature operating point or an engineering criterion, not as a mathematically resolved threshold. This frequency scan uses the static effective pair drive, whose Hamiltonian is time-independent, so the phase of $\langle \hat{a}\hat{a} \rangle$ relaxes to a constant and the scan does not define a within-run phase statistic: the circular standard deviations quoted in Section 6 and Table 2 of the main text come from separate runs of the resonantly modulated phase-control protocol and must not be read off this scan. What the scan does establish for pair-phase encoding is the thermal input to those runs, the occupation $\bar{n}_{\text{th}}$ of the driven mode, which falls from 128.7 at 50 GHz to 0.65 at 6 THz.

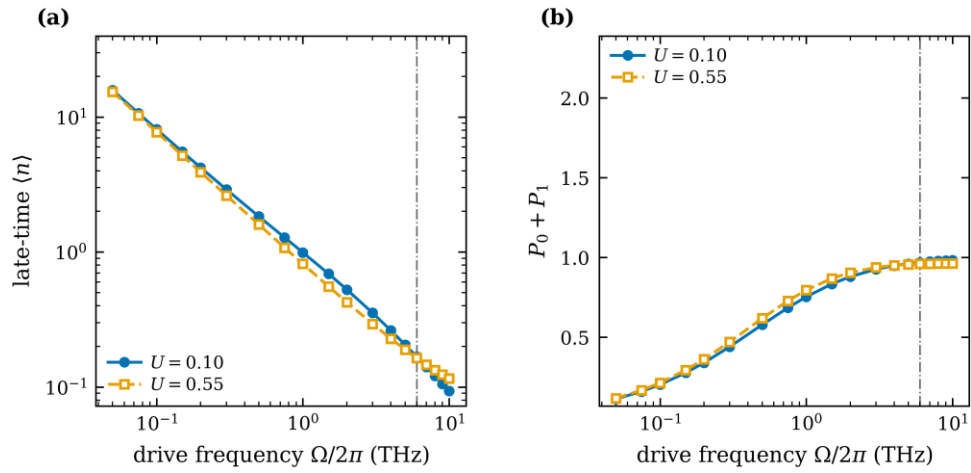


Figure S4. Finite-temperature frequency scan at 310 K. Final mean occupation and low-manifold population are shown for $U_{\text{audit}} = 0.10$ and $U_{\text{audit}} = 0.55$; leakage and pair-correlation measures are shown for $U_{\text{audit}} = 0.55$. The frequency axis is logarithmic. The static effective pair-drive scan demonstrates progressive thermal recovery with frequency and no singular threshold at 6 THz.

Supplement 9. Finite-temperature protocol: initial state, relaxation time, and detailed-balance validity

Two finite-temperature protocols are used in this work and are kept separate. A calculation that starts from the bosonic vacuum and stops at $\Omega t = 100$ ($\kappa t = 0.2$ for $\alpha = 10^{-3}$) describes transient heating after an idealized reset rather than a thermal operating state; all $P_0 + P_1$ values quoted in the main text are therefore obtained from thermal-start calculations integrated to $\kappa t = 3$ (S9.1–S9.5). The high-occupation phase-statistics campaign starts from the vacuum and is used as an explicitly labelled reset-and-read transient phase-response calculation. It underlies the room-temperature 50-GHz phase route discussed in the main text, but it is not a thermal steady-state phase-error calculation. Its Fock-basis convergence must therefore be judged on the transient state actually simulated, not on the equilibrium thermal tail; that separate audit is given in S9.7.

S9.1 Thermal-start protocol

The thermal-start calculation starts from the bath's own thermal state (QuTiP `thermal_dm` at the bare-mode occupation $\bar{n}_{\text{th}}$), uses the same resonantly modulated Hamiltonian and thermal collapse operators as Supplement 8, and integrates to $\kappa t = 3$, several damping times past the initial transient, rather than to $\Omega t = 100$. The Fock basis is sized from the tail-bound formula above with a safety margin for the drive-induced occupation, and convergence is checked by verifying that the population in the top Fock levels is negligible at the final time.

S9.2 Results for Regimes I and III

Regime I ($T = 4$ K, 50 GHz, $\bar{n}_{\text{th}} = 1.2166$, $N = 25$, exact Lindblad propagation): late-time mean occupation $\langle n \rangle = 1.936$, $P_0 = 0.2428$, $P_1 = 0.2533$, $P_0 + P_1 = 0.496$, $P(n \geq 2) = 0.504$. This differs from the $P_0 + P_1 \approx 0.95$ of the zero-temperature static benchmark of Figs. 1–3 of the main text, which uses a different drive protocol and neglects thermal injection.

Regime III ($T = 310$ K, 6 THz, $\bar{n}_{\text{th}} = 0.6529$, $N = 25$, exact Lindblad propagation): late-time mean occupation $\langle n \rangle = 1.611$, $P_0 = 0.2834$, $P_1 = 0.3005$, $P_0 + P_1 = 0.584$. This differs from the value 0.925 of the static-drive frequency scan of Supplement 8, which uses a different drive protocol and a 20-pump-period vacuum start.

In both cases the pair-phase circular standard deviation σ (22.4° for Regime I, 26.1° for Regime III) is reproduced independently under the thermal-start protocol.

S9.3 Detailed-balance validity of the single-frequency thermal dissipator

The thermal dissipator $\kappa(\bar{n}_{\text{th}} + 1)\mathcal{D}[\hat{a}]\rho + \kappa\bar{n}_{\text{th}}\mathcal{D}[\hat{a}^\dagger]\rho$ used throughout this work evaluates a single Bose occupation $\bar{n}_{\text{th}}$ at the bare mode frequency Ω for every transition. This is exact for a harmonic spectrum but an approximation for the Kerr-shifted spectrum used here, where adjacent-level transition frequencies are $\omega(n \rightarrow n + 1) = \Omega(1 + U_{\text{audit}}n)$, so each additional quantum shifts the transition frequency by $\Delta f_{\text{Kerr}} = U_{\text{audit}}f$ relative to the bare frequency $f = \Omega/2\pi$. Using a single bare-frequency $\bar{n}_{\text{th}}$ for every transition is justified when this shift is small compared with the thermal frequency scale $k_B T/h$, that is, when the dimensionless ratio $r = \Delta f_{\text{Kerr}}/(k_B T/h) = hU_{\text{audit}}f/(k_B T)$ is small. At $U_{\text{audit}} = 0.55$ this ratio is $r = 0.0043$ for Regime IV (310 K, 50 GHz) and $r = 0.017$ for Regime II (77 K, 50 GHz), so the single-frequency dissipator is well justified for both; it is $r = 0.33$ for Regime I (4 K, 50 GHz) and $r = 0.51$ for Regime III (310 K, 6 THz), where the approximation is marginal. These are also the two regimes in which the resonant thermal-start results of S9.2 depart most strongly from the static benchmarks, consistent with a common cause: at these two operating points the Kerr anharmonicity is comparable to the relevant thermal scale, so approximations that treat it as a small correction are the ones most likely to break down. A transition-resolved Kerr-basis dissipator, using each transition's own occupation $\bar{n}_{\text{th}}[\omega(n \rightarrow n + 1)]$ rather than a single bare-frequency value, addresses this approximation within the static Kerr ladder for Regimes I and III; S9.6 reports that check. A full Floquet-resolved master equation would be a separate extension.

S9.4 Thermal baseline and the unavailability of a linear QLE benchmark

Two further consistency checks were performed. First, a purely thermal (undriven) mode at the same $\bar{n}_{\text{th}}$ has a definite low-manifold population, $P_0 + P_1 = 1/(1 + \bar{n}_{\text{th}}) + \bar{n}_{\text{th}}/(1 + \bar{n}_{\text{th}})^2$, giving a baseline of 0.6987 (Regime I), 0.0604 (Regime II), 0.8440 (Regime III) and 0.0154 (Regime IV). The driven values of S9.2 fall below this

baseline for Regimes I and III (0.496 and 0.584, against 0.699 and 0.844), while the Regime II trajectory estimate sits above its much smaller baseline. For Regime IV the eight-trajectory thermal-start sample does not resolve P_0+P_1 at this level (S9.5).

Second, we examined a companion linear ($U = 0$) second-moment QLE model with the same resonant Hamiltonian coefficient used in the numerical protocol, $g(t) = (\eta\Omega/2) \sin(2\Omega t + \varphi)$. The exact noise-included equations are $dn/dt = -\kappa n + \kappa \bar{n}_{\text{th}} - 4g(t) \text{Im } m$ and $dm/dt = -(\kappa + 2i\Omega)m - 2ig(t)(2n + 1)$, where $n = \langle \hat{a}^\dagger \hat{a} \rangle$ and $m = \langle \hat{a} \hat{a} \rangle$. The linear model diverges already at the smaller envelope $\eta\Omega/4$, and the protocol-matched coefficient $\eta\Omega/2$ can only increase the linear gain. Thus no bounded linear steady-state benchmark exists at $\eta = 1.2$ and $\alpha = 10^{-3}$: the operating point lies far above the linear parametric threshold $\eta > 2\alpha$, and the Kerr term is the mechanism that supplies boundedness. This QLE argument is used only to establish the absence of a bounded $U = 0$ reference, not to generate the nonlinear phase statistics.

S9.5 High-occupation trajectory calculations: protocol and convergence

Regime II ($T = 77$ K, 50 GHz, $\bar{n}_{\text{th}} = 31.6$) and Regime IV ($T = 310$ K, 50 GHz, $\bar{n}_{\text{th}} = 129$) require substantially larger Fock bases than Regimes I and III. Regime II was evaluated under the thermal-start protocol (thermal initial state, $\kappa t = 3$) with quantum-trajectory integration at $N = 220$. The 56-trajectory estimate gives $P_0+P_1 = 0.107 \pm 0.045$, establishing that Fock-manifold confinement is not maintained at 77 K under the resonant drive. For Regime IV the same thermal-start protocol is evaluated at $N = 550$ using segmented quantum-trajectory integration with the drive phase propagated in absolute time across segment boundaries. Eight trajectories were integrated, each started from a Fock state sampled from the thermal distribution (initial occupations 4–249) and propagated to $\kappa t = 3$ in segments of $\kappa t = 0.1$. The late-time mean occupation is $\langle n \rangle = 140 \pm 37$ (standard error over trajectories; individual trajectories 30–328), consistent with the bath occupation $\bar{n}_{\text{th}} = 129$; the thermal-only weight beyond the basis is $P(n \geq 550) = 0.014$. Seven of the eight trajectories end with $P_0+P_1 = 0$ to machine precision and one with 3.7×10^{-8} , so the sample does not resolve the low-manifold population: the undriven thermal baseline 0.0154 (S9.4) corresponds to about one trajectory in 65 near the bottom of the ladder, and resolving a value of this size would require several hundred trajectories. The result is therefore reported in Table 2 of the main text as not resolved (n.r.); what it establishes is that the driven mode sits at the bath occupation, with no Fock-manifold confinement. The separate reset-and-read transient calculation remains the basis for the room-temperature 50-GHz phase result; its basis convergence is documented in S9.7.

S9.6 Transition-resolved Kerr-basis master-equation check for Regimes I and III

S9.3 (main text Section 6) estimates, via the dimensionless ratio $r = \hbar U_{\text{audit}}/f(k_B T)$, that the single bath-occupation dissipator $\kappa(\bar{n}_{\text{th}} + 1)\mathcal{D}[\hat{a}]\rho + \kappa\bar{n}_{\text{th}}\mathcal{D}[\hat{a}^\dagger]\rho$ is marginal for Regimes I ($r = 0.33$) and III ($r = 0.51$). This subsection reports a direct transition-resolved Kerr-basis dissipator comparison rather than leaving r as an unverified sensitivity indicator. It is not a full Floquet master-equation treatment.

Construction. The Kerr-ladder energies are $E_n = \Omega n + (U_{\text{audit}}\Omega/2)n(n-1)$, $n = 0, \dots, N-1$, giving adjacent-level Bohr frequencies $\omega_n = E_n - E_{n-1} = \Omega[1 + U_{\text{audit}}(n-1)]$. For each transition $n \leftrightarrow n-1$ the physical frequency is $f_n = f_{\text{regime}}\omega_n$, and the corresponding thermal occupation is $\bar{n}_{\text{th}}(f_n) = 1/[\exp(\hbar f_n/k_B T) - 1]$. The single-operator dissipator, with jump operators $[\kappa(\bar{n}_{\text{th}} + 1)]^{1/2}\hat{a}$ and $(\kappa\bar{n}_{\text{th}})^{1/2}\hat{a}^\dagger$, is replaced by $2(N-1)$ transition-specific jump operators: for each n from 1 to $N-1$, $[\kappa(\bar{n}_{\text{th}}(f_n) + 1)]^{1/2}n^{1/2}|n-1\rangle\langle n|$ (emission) and $[\kappa\bar{n}_{\text{th}}(f_n)]^{1/2}n^{1/2}|n\rangle\langle n-1|$ (absorption). The initial state is also changed from the bare-harmonic thermal state to the anharmonic Gibbs state, with weights proportional to $\exp(-E_n\hbar f_{\text{regime}}/k_B T)$, built from the Kerr ladder's own energies rather than assuming a harmonic spectrum. With the drive switched off, each transition obeys $\Gamma_\uparrow/\Gamma_\downarrow = \bar{n}_{\text{th}}(\omega_n)/[\bar{n}_{\text{th}}(\omega_n) + 1] = \exp(-\beta\hbar\omega_n)$; consequently the anharmonic Gibbs weights satisfy pairwise detailed balance and are stationary under this transition-resolved dissipator.

Results ($N = 25$, $\kappa t = 3$, and the same $U_{\text{audit}} = 0.55$, $\eta = 1.2$ and $\alpha = 10^{-3}$ as in the rest of this section). Regime I (4 K, 50 GHz): $P_0+P_1 = 0.433$ under the transition-resolved dissipator, against 0.496 for the single-frequency treatment used in Table 2, a 13% reduction. Regime III (310 K, 6 THz): $P_0+P_1 = 0.484$ transition-resolved, against 0.584 single-frequency, a 17% reduction.

Isolating the two changes (transition-resolved dissipator versus anharmonic Gibbs initial state) shows that the initial-state change alone shifts P_0+P_1 by under 0.3% in both regimes (Regime I: 0.496 to 0.497; Regime III: 0.584 to 0.584); essentially all of the 13–17% shift comes from the dissipator, not from the initial condition. Both refinements reduce P_0+P_1 further, that is, they point away from confinement, consistent with the direction found for the resonant thermal-start protocol in S9.2. The relative size of the shift (17% for Regime III against 13% for Regime I) tracks the ranking of r (0.51 against 0.33) established analytically in S9.3, so the two independent checks — the analytical sensitivity ratio and this direct numerical comparison — agree in both sign and relative magnitude.

For Regimes I and III the single-frequency values are retained in Table 2 for methodological continuity, with the 13–17% transition-resolved shift quantified here as a sensitivity check. Regimes II and IV are already in the small- r regime, so this transition-resolved correction is not required for their population results. The 13–17% percentages apply only to P_0+P_1 in this check and are not general error bounds on other observables.

S9.7 Reset-and-read transient phase response: protocol scope and basis convergence

The phase σ and BER columns of Table 2 come from a separate idealized reset-and-read diagnostic, not from the thermal-start population calculation. The phase runs start from the vacuum, turn on the thermal bath at the listed temperature together with the resonant drive, use $U_{\text{audit}} = 0.55$ and $\alpha = 10^{-3}$, and are followed to $\Omega t = 100$. The circular standard deviation is computed from the rotating-frame phase of $\langle \hat{a}\hat{a} \rangle$ over the second half of that transient. This is the transient-heating protocol described at the start of this Supplement; no thermal steady-state σ or BER is inferred for Regimes II or IV. Regimes I and III were independently rerun from the thermal state to $\kappa t = 3$ and reproduce 22.4° and 26.1° , respectively.

Basis convergence for the reset transient is direct. The values below are from the archived phase-statistics campaign; “top-10 max” is the maximum population carried by the highest ten retained Fock levels during the run. $N = 240$ would be inadequate for an equilibrium $\bar{n} = 129$ state, but the diagnostic does not represent such a state. For the transient observable actually reported, σ changes by only 0.0072° from $N = 200$ to 240 while the top-10 population falls to 5.0×10^{-5} . Thus the 40.7° figure is numerically converged as a reset-and-read transient phase-response observable. It supports Regime IV as a room-temperature microwave phase route within that protocol, not as an equilibrium phase-noise or continuous steady-state error-rate result.

Regime	N	σ (deg)	$\Delta\sigma$ from previous N (deg)	top-10 max
I	50	22.44025	—	3.67×10^{-23}
I	80	22.43986	0.00039	2.66×10^{-43}
II	80	29.93584	—	4.34×10^{-5}
II	120	29.94798	0.01214	1.06×10^{-7}
III	50	26.14145	—	1.65×10^{-25}
III	80	26.14069	0.00076	2.77×10^{-52}
IV	160	40.66185	—	1.30×10^{-3}
IV	200	40.69786	0.03601	2.54×10^{-4}
IV	240	40.70504	0.00718	4.96×10^{-5}

Supplement 10. Quadratic anisotropy decomposition and scope of the pair-drive mapping

This supplement separates an exact algebraic identity from the additional physical assumptions required to map an experimental anisotropy modulation onto the effective pair-drive coefficient used numerically. The quadratic decomposition is exact, but it does not make the numerical pump amplitude a unique function of a scalar instantaneous frequency.

S10.1 Laboratory-frame transverse-curvature Hamiltonian

For a small-angle magnetic mode, let $\hat{X} = (\hat{a} + \hat{a}^\dagger)/\sqrt{2}$ and $\hat{P} = (\hat{a} - \hat{a}^\dagger)/(i\sqrt{2})$ be fixed canonical quadratures. Write the most general diagonal quadratic transverse-anisotropy form as $\hat{H}_q(t) = (\hbar/2)[\Omega_x(t)\hat{X}^2 + \Omega_y(t)\hat{P}^2]$, where Ω_x and Ω_y are the two instantaneous transverse energy curvatures about the reference axis.

S10.2 Exact fixed-basis decomposition

Substituting the fixed quadratures gives exactly $\hat{H}_q(t) = \hbar A(t)(\hat{n} + 1/2) + \hbar \mathcal{G}(t)(\hat{a}^{\dagger 2} + \hat{a}^2)$, with $A(t) = [\Omega_x(t) + \Omega_y(t)]/2$ and $\mathcal{G}(t) = [\Omega_x(t) - \Omega_y(t)]/4$. The physical small-oscillation eigenfrequency of the quadratic form is $\omega_{\text{inst}}(t) = [\Omega_x(t)\Omega_y(t)]^{1/2} = [A(t)^2 - 4\mathcal{G}(t)^2]^{1/2}$. Therefore $A(t)$ is the diagonal coefficient in the chosen fixed basis, but it is not generally identical to the physical eigenfrequency when $\mathcal{G}$ is appreciable.

S10.3 What the decomposition does and does not determine

The exact identities above show that a laboratory anisotropy modulation controls two independent combinations: the sum of the transverse curvatures sets $A(t)$, while their difference sets $\mathcal{G}(t)$. Consequently, specifying a scalar frequency trajectory, or the sweep metric $\eta_{\text{NA}}(t) = |d\Omega/dt|/\omega^2$ by itself, is insufficient to determine the pair coefficient $\mathcal{G}(t)$. Conversely, choosing a curvature path solely to reproduce a desired $\mathcal{G}(t)$ generally also changes $A(t)$ and the physical eigenfrequency. For this reason the manuscript does not identify the numerical pair-pump amplitude with a unique laboratory frequency sweep.

The numerical resonant Hamiltonian is therefore defined directly at the effective-mode level by $\mathcal{G}_{\text{num}}(t) = (\eta\Omega/2) \sin(2\Omega t + \varphi)$, while the static benchmark uses $\mathcal{G}_{\text{static}} = \eta\Omega/4$. In these expressions η is the effective pump-amplitude parameter. Equation (3) of the main text defines the separate local sweep quantity η_{NA} . A material-specific relation between η and η_{NA} would require a microscopic model of how the optical or microwave control changes the two transverse anisotropy curvatures.

Static protocol. The confinement scan uses the separate constant Hamiltonian coefficient $\mathcal{G}_{\text{static}} = \eta\Omega/4$. Because the diagonal mode coefficient is held fixed in that benchmark, the derivative-based sweep non-adiabaticity of Eq. (3) is not what labels this calculation; η there is only the effective drive-amplitude parameter.

S10.4 Scope and approximation status

The fixed-basis decomposition $A = (\Omega_x + \Omega_y)/2$ and $\mathcal{G} = (\Omega_x - \Omega_y)/4$ is algebraically exact. No rotating-wave approximation is required to write the numerical effective Hamiltonian itself. The unresolved step is different: the present work does not derive the experimentally induced functions $\Omega_x(t)$ and $\Omega_y(t)$ from a microscopic optical or microwave pump, so it does not provide a unique laboratory calibration between η_{NA} and the effective numerical pump amplitude η . This limitation is stated explicitly rather than absorbed into a convention choice.

The fixed-basis identities were cross-checked symbolically. They establish the exact algebraic relation between transverse curvatures and the diagonal/pair coefficients, while also making clear that one scalar frequency trajectory cannot by itself determine both coefficients.